

CLASS 9 MATHEMATICS

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CHAPTER 4 – EXPLORING ALGEBRAIC IDENTITIES

EXERCISE SET 4.4

IMPORTANT IDENTITIES

$$(a + b)(a - b) = a^2 - b^2$$

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

1. Fill in the blanks to complete the following identities:

(i) $s^2 - 11s + 24 = (\text{_____})(\text{_____})$

(ii) $(\text{_____})(x + 1) = (3x^2 - 4x - 7)$

(iii) $10x^2 - 11x - 6 = (2x - \text{_____})(\text{_____} + 2)$

(iv) $6x^2 + 7x + 2 = (\text{_____})(\text{_____})$

Solution

(i)

$$s^2 - 11s + 24$$

We need two numbers whose sum is -11 and product is 24 .

These numbers are -3 and -8 .

Therefore,

$$s^2 - 11s + 24 = (s - 3)(s - 8).$$

Hence,

$$(s - 3)(s - 8)$$

(ii)

We have

$$(\text{_____})(x + 1) = 3x^2 - 4x - 7.$$

Factorising the quadratic,

$$3x^2 - 4x - 7 = (3x - 7)(x + 1).$$

Therefore,

$$\boxed{(3x - 7)(x + 1)}$$

So the blank is

$$\boxed{3x - 7}.$$

(iii)

$$10x^2 - 11x - 6$$

We need factors of $10x^2$ and -6 .

$$10x^2 - 11x - 6 = (2x - 3)(5x + 2).$$

Checking,

$$(2x - 3)(5x + 2)$$

$$= 10x^2 + 4x - 15x - 6$$

$$= 10x^2 - 11x - 6.$$

Therefore,

$$\boxed{(2x - 3)(5x + 2)}$$

So the blanks are

$$\boxed{3} \quad \text{and} \quad \boxed{5x}.$$

(iv)

$$6x^2 + 7x + 2$$

Factorising,

$$6x^2 + 7x + 2 = (3x + 1)(2x + 2).$$

Checking,

$$(3x + 1)(2x + 2)$$

$$= 6x^2 + 6x + 2x + 2$$

$$= 6x^2 + 7x + 2.$$

Hence,

$$\boxed{(3x + 1)(2x + 2)}$$

2. Select and use the identity that will help you to find the following products without multiplying directly:

(i) $(41)^2$

(ii) $(27)^2$

(iii) (23×17)

(iv) $(135)^2$

(v) $(97)^2$

(vi) (18×29)

(vii) (34×43)

(viii) $(205)^2$

Solution

(i)

$$41^2 = (40 + 1)^2$$

Using

$$(a + b)^2 = a^2 + 2ab + b^2,$$

$$= 40^2 + 2(40)(1) + 1^2$$

$$= 1600 + 80 + 1$$

$$\boxed{41^2 = 1681}$$

(ii)

$$27^2 = (30 - 3)^2$$

Using

$$(a - b)^2 = a^2 - 2ab + b^2,$$

$$= 30^2 - 2(30)(3) + 3^2$$

$$= 900 - 180 + 9$$

$$\boxed{27^2 = 729}$$

(iii)

$$23 \times 17$$

Write

$$23 = 20 + 3, \quad 17 = 20 - 3.$$

Therefore,

$$23 \times 17 = (20 + 3)(20 - 3).$$

Using

$$(a + b)(a - b) = a^2 - b^2,$$

$$= 20^2 - 3^2$$

$$= 400 - 9$$

$$\boxed{23 \times 17 = 391}$$

(iv)

$$135^2 = (100 + 35)^2$$

Using

$$(a + b)^2 = a^2 + 2ab + b^2,$$

$$= 100^2 + 2(100)(35) + 35^2$$

$$= 10000 + 7000 + 1225$$

$$\boxed{135^2 = 18225}$$

(v)

$$97^2 = (100 - 3)^2$$

$$= 100^2 - 2(100)(3) + 3^2$$

$$= 10000 - 600 + 9$$

$$\boxed{97^2 = 9409}$$

(vi)

$$18 \times 29$$

Write

$$18 = 20 - 2, \quad 29 = 30 - 1.$$

Thus,

$$18 \times 29 = (20 - 2)(30 - 1).$$

Expanding,

$$= 600 - 20 - 60 + 2$$

$$= 522.$$

Therefore,

$$\boxed{18 \times 29 = 522}$$

(vii)

$$34 \times 43$$

Write

$$34 = 30 + 4, \quad 43 = 40 + 3.$$

Therefore,

$$34 \times 43 = (30 + 4)(40 + 3).$$

$$= 1200 + 90 + 160 + 12$$

$$= 1462.$$

Hence,

$$\boxed{34 \times 43 = 1462}$$

(viii)

$$205^2 = (200 + 5)^2$$

Using

$$(a + b)^2 = a^2 + 2ab + b^2,$$

$$= 200^2 + 2(200)(5) + 5^2$$

$$= 40000 + 2000 + 25$$

$$\boxed{205^2 = 42025}$$

3. Factor the following:

(i) $9a^2 + b^2 + 4c^2 - 6ab + 12ac - 4bc$

(ii) $16s^2 + 25t^2 - 40st$

(iii) $r^2 - r - 42$

(iv) $49g^2 + 14gh + h^2$

(v) $64u^2 + 121v^2 + 4w^2 - 176uv - 32uw + 44vw$

(i)

$$9a^2 + b^2 + 4c^2 - 6ab + 12ac - 4bc$$

Observe that

$$9a^2 = (3a)^2, \quad b^2 = (-b)^2, \quad 4c^2 = (2c)^2.$$

Also,

$$2(3a)(-b) = -6ab,$$

$$2(3a)(2c) = 12ac,$$

$$2(-b)(2c) = -4bc.$$

Therefore,

$$\begin{aligned} 9a^2 + b^2 + 4c^2 - 6ab + 12ac - 4bc \\ = (3a - b + 2c)^2. \end{aligned}$$

Hence,

$$\boxed{(3a - b + 2c)^2}$$

(ii)

$$16s^2 + 25t^2 - 40st$$

We have

$$16s^2 = (4s)^2, \quad 25t^2 = (5t)^2.$$

Also,

$$2(4s)(5t) = 40st.$$

Therefore,

$$\begin{aligned} &16s^2 + 25t^2 - 40st \\ &= (4s)^2 - 2(4s)(5t) + (5t)^2 \end{aligned}$$

$$\boxed{(4s - 5t)^2}$$

(iii)

$$r^2 - r - 42$$

We need two numbers whose product is -42 and whose sum is -1 .

These numbers are -7 and 6 .

Therefore,

$$r^2 - r - 42 = (r - 7)(r + 6).$$

Hence,

$$\boxed{(r - 7)(r + 6)}$$

(iv)

$$49g^2 + 14gh + h^2$$

We have

$$49g^2 = (7g)^2, \quad h^2 = h^2.$$

Also,

$$2(7g)(h) = 14gh.$$

Therefore,

$$49g^2 + 14gh + h^2 = (7g + h)^2.$$

Hence,

$$\boxed{(7g + h)^2}$$

(v)

$$64u^2 + 121v^2 + 4w^2 - 176uv - 32uw + 44vw$$

Observe that

$$64u^2 = (8u)^2,$$

$$121v^2 = (-11v)^2,$$

and

$$4w^2 = (-2w)^2.$$

Also,

$$2(8u)(-11v) = -176uv,$$

$$2(8u)(-2w) = -32uw,$$

and

$$2(-11v)(-2w) = 44vw.$$

Therefore,

$$\begin{aligned} 64u^2 + 121v^2 + 4w^2 - 176uv - 32uw + 44vw \\ = (8u - 11v - 2w)^2. \end{aligned}$$

Hence,

$$(8u - 11v - 2w)^2$$

Think and Reflect

James and Reshma were talking about algebraic identities they learnt in school.

James:

$$(a - b)^2(a + b) = (a^2 - 2ab + b^2)(a + b)$$

Reshma:

$$\begin{aligned} (a - b)^2(a + b) &= (a - b)[(a - b)(a + b)] \\ &= (a - b)(a^2 - b^2). \end{aligned}$$

Question:

According to you, who is correct and why?

Try to combine more such identities and find new results.

Solution

Both James and Reshma are correct.
Starting with

$$(a - b)^2(a + b),$$

we can use

$$(a - b)^2 = a^2 - 2ab + b^2$$

to get James' form:

$$\boxed{(a^2 - 2ab + b^2)(a + b)}.$$

Alternatively, group the factors as

$$(a - b)^2(a + b) = (a - b)(a - b)(a + b).$$

Now use

$$(a - b)(a + b) = a^2 - b^2.$$

Therefore,

$$\boxed{(a - b)(a^2 - b^2)}.$$

Hence, both forms are equivalent.

EXERCISE 4.4 – QUICK REVISION

$$(a + b)(a - b) = a^2 - b^2$$

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

For factorisation:

$$x^2 + (a + b)x + ab = (x + a)(x + b)$$